## Unit 3:-

**Topics:** Basic theory of Laplace Transform, Laplace Transform of elementary functions, Inverse Laplace Transform

Problems:

- Example 8.1-8.5 (on page 8.2 & 8.3) (J&I)
- Q. 1- 10/Exercise 8.1 (on page 8.6) (J&I)
- Q. 19—22/Exercise 8.1 (on page 8.6) (J&I)
- Q. 31—40/Exercise 8.1 (on page 8.6 & 8.7) (J&I)

**Topics:** Laplace Transform of derivatives, Laplace Transform of Integrals, Translation Theorem (First Shifting Property)

Problems:

- Example 8.16, 8.17 & 8.22 (on page 8.11, 8.12 & 8.15) (J&I)
- Q. 1—19/Exercise 8.2 (on page 8.13) (J&I)
- Q. 1—21/Exercise 8.3 (on page 8.22) (J&I)

**Topics:** Differentiation of Laplace Transform, Integration of Laplace Transform, Unit Step Functions, Translation Theorem (Second Shifting Property)

Problems:

- Example 8.32, 8.33 & 8.37 (on page 8.29 & 8.33) (J&I)
- Q. 10—22/Exercise 8.4 (on page 8.38) (J&I)
- Q. 32—42/Exercise 8.4 (on page 8.38) (J&I)
- Q. 29—52/Exercise 8.3 (on page 8.23) (J&I)

**Topics:** Periodic Functions, Convolution Theorem

Problems:

- Q. 2, 3, 6, 8 /Exercise 8.5 (on page 8.43) (J&I)
- Q. 46—55/Exercise 8.4 (on page 8.38) (J&I)

**Topic:** Applications to linear differential equations (Initial Value Problems)

Problems:

- Q. 23—36/Exercise 8.2 (on page 8.13) (J&I)
- Q. 22—28/Exercise 8.3 (on page 8.22)
- Q. 23—31/Exercise 8.4 (on page 8.38) & Q. 58—64/Exercise 8.4 (on page 8.39) (J&I)
- Q. 53—65/Exercise 8.3 (on page 8.24) (J&I)
- Q. 14—20/Exercise 8.5 (on page 8.44) (J&I)

(Dr. Ghazala Yasmin)
Course Incharge